Math 126

Extra Unit C practice

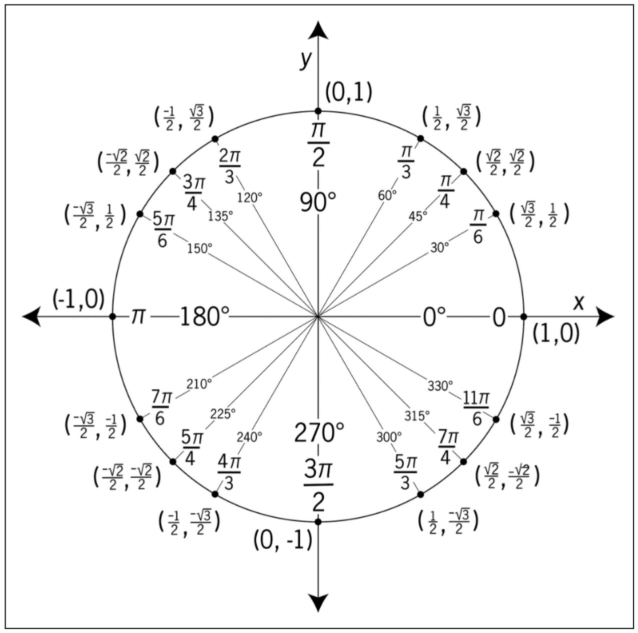
The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity C.

Geometric series: $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \ |r| < 1$

Alternating Series: $\left| s_n - \sum_{i=0}^{\infty} (-1)^i b_i \right| \leq b_{n+1}$

Taylor series coefficients: $c_n = \frac{f^{(n)}(a)}{n!}$

Taylor's Formula: $R_n(x) = \frac{f^{(n+1)}(z)}{n!}(x-a)^{n+1}$



Maclaurin series

Series	R
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$	1
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	∞
$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	∞
$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	∞
$\tan^{-1}(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$	1
$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	1
$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \cdots$	1

C1 - Sequences

(a) Determine whether the sequence $a_n = \dfrac{(-2)^n}{n!}$ converges or diverges. Explain or show your reasoning.

(b) Determine whether the sequence $b_n = \cos\left(\dfrac{n\pi}{2}\right)$ converges or diverges. Explain or show your reasoning.

(c) Consider the sequence defined recursively via $c_1 = 2$ and

$$c_{n+1} = \frac{1}{3 - c_n}$$

You may assume that this sequence is decreasing.

(i) Show that this sequence converges. Clearly explain your reasoning.

(ii) Find $\lim_{n \to \infty} c_n$.

C2 - Series

(a) Suppose that the n^{th} partial sum of a series is $s_n = 5 - \dfrac{n^2}{3^{n-1}}$. Notice that $a_1 = s_1 = 4$.

(i) Find a formula for a_n when $n > 1$. You do not need to simplify your formula.

(ii) Find $\sum_{n=1}^{\infty} a_n$.

(b) Determine the sum of the series $\sum_{n=3}^{\infty} \dfrac{4}{n^2 - 4}$

C3 - Convergence tests

Determine whether each of the following series converges or diverges. State the names of any tests you use.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin^2(n)}{n^3 + 2n^2 - 1}$

(b) $\displaystyle\sum_{n=1}^{\infty} ne^{-n}$

(c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{n}{2n+1}$

(d) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(\frac{n}{2n+1}\right)^n$

C4 - Power series

(a) Find the radius of convergence of the series $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$

(b) Let $f(x) = \frac{x^2}{1+x^7}$

(i) Find a power series representation of $f(x)$.

(ii) Find a power series representation of $\int f(x)dx$.

C5 - Taylor series

(a) Evaluate the indefinite integral $\int \frac{\sin(x) - x}{x^3} dx$ as an infinite series.

(b) Find the second degree Taylor polynomial of $f(x) = x^{\frac{4}{3}}$ at $a = 8$ and find the largest possible error of this approximation on the interval $6 \leq x \leq 10$.